Zero human in the inner loop

Position note on why Pathfinder exists

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1 The argument

Assume that AI agents built on the latest language models soon match or surpass humans at solving and checking most scientific and technical tasks. Both in cost and speed of execution. Arguably, this is already the case in mathematics (Navier-Stokes). Then the scarce resource in research is no longer to solve problems but to formulate good ones. The main focus shifts from problem solving to idea generation.

It is often said these days that humans cannot understand new math results at the speed agents produce them. Assume here that human understanding is valuable because it lends eventually to new ideas. A solve → publish → understand → new-idea-formation cycle. Hereafter the "inner loop". If human understanding is a necessary ingredient of the inner loop, then AI will not accelerate science. Because humans cannot produce new ideas/questions at the speed agents solve them. Assume human understanding is not.

[experiment: ask a strong AI agent if it understands a recent proof from the dump]

The human contribution therefore moves from producing ideas to designing the machines that produce them. Engineering agent systems that generate scientific ideas becomes the most important task in science, at least until agents can design such systems themselves.

[generating a problem even a difficult one is easy, just take large random instances of 3SAT near criticality; so here we want to generate ideas rather than meaningless problems]

Pathfinder (`https://github.com/vd1/pathfinder`) is a bet on this trend rather than a description of today: in its first campaign (nobel) its ideas held up better than its solutions, which still fall short of human papers on the same question. It runs with no human intervention, because human clocks are much slower than machine ones and any human step in the loop becomes its bottleneck.

2 Three qualifications

Zero intervention means the inner loop. No human acts during a run, but humans choose the corpus, the rater, its anchor scale and the prompts. In nobel, most of the real gains came from corrections at that level: reading sources as LaTeX instead of converted PDFs, validating the rater, adding a presentation pass. This is the "designing the machines" role of the argument, and it means human time still matters, in the outer loop, which is where Pathfinder has improved so far. The principle is therefore no human in the inner loop, not no human at all.

The bottleneck may be evaluation rather than generation. Even with strong solvers, a system that generates ideas needs a judge of which ideas matter. Today that judge is a language model rater anchored to human taste. A rater that tracks current consensus will find good ideas

near the frontier but is unlikely to reward the ideas that move it. A large part of "agents engineering their own idea discovery systems" is building a judge that is not anchored to people.

Solving and generating may not separate cleanly. Nobel's weakest result was execution: about 2.5 points below the human papers realising the same idea, on the campaign's rating scale. Many good questions only appear while pushing on a hard problem, which is the intuition behind turning obstacles into subgoals in the growth solver. If generation and solving are coupled, generation cannot run far ahead of solving, and getting solving right remains part of the idea generation project rather than something to hand off.

3 Other ways to generate ideas

Pathfinder draws novelty from collision: two papers that do not cite each other, read together, yield a question neither asks. Other generators draw it from elsewhere.

From the literature, beyond pairs.

- *Gaps in a single paper.* Mine a strong paper's conjectures, open problems and "it would be interesting to" sentences, and the hypotheses its proofs use but never question (can this one be dropped or weakened?). Cheap, and the questions arrive with community relevance attached.
- *Analogy transfer.* Extract the abstract shape of a theorem or technique, then search for fields where that shape has no counterpart yet. Pathfinder does this implicitly through pairs; here it is done on purpose.
- *Citation graph holes.* Find clusters that share objects or methods but do not cite each other. This chooses which pairs to collide rather than scanning pairs.
- *Negative space.* Find the generalisation everyone would expect and check why it is missing: false, hard, or overlooked. Each answer is a different kind of question.

From doing, not reading.

- *Questions from failed proofs.* The growth solver intuition taken as the main source: attack a hard known problem and record every obstacle as a candidate question. The ideas come motivated (they unblock something), and the generator addresses the coupling of solving and generating directly.
- *Experiment first.* Compute many small cases (in the style of AlphaEvolve, FunSearch or the Ramanujan Machine), look for unstated patterns, then conjecture. Limited to computable domains, but there the ideas rest on evidence rather than plausibility.
- *Formal libraries.* A library at the scale of Mathlib exposes missing lemmas, definitions without an interface, and statements that almost go through. A plain source, but every idea is precise.

From changing the frame.

- *Pólya operators applied mechanically.* Generalise, specialise, drop a hypothesis, swap quantifiers, dualise, make the qualitative quantitative, make the existential constructive. Apply them to known theorems and rate the variants. Pathfinder's question formation could use this as a second generator.

- *Reverse from goals.* Start from a flagship open problem or an application need, decompose it backwards into intermediate statements, and keep the ones that are new and look reachable.
- *Outside demand.* Translate bottlenecks named by engineers, experimentalists or other sciences into theory questions. This supplies relevance rather than novelty.

From the judge side.

- *Generate wide, select hard.* Produce questions in bulk with any of the above and put the effort into the judge: tournaments, predicted impact, the test "would a top expert be surprised?". Generation is cheap; judging is the limit.
- *Open-ended archives.* Novelty search in the MAP-Elites style: keep questions in an archive over descriptors (field, type, difficulty) and reward filling empty cells rather than raising the top score. This counters the pull of a rater anchored to consensus.

Two of these complement Pathfinder best, and both are cheap to test with the nobel machinery: questions from failed proofs, which feeds the solving to generation path that nobel found weak, and generate wide with an archive, which targets the judge. Collision, Pólya operators and gap mining could then feed one archive and one judge, and the comparison would show which generator yields the best ideas.

4 Consequence for Pathfinder

None of the qualifications changes the conclusion. They change where to put the effort inside Pathfinder: the judge, and the path from solving back into question formation.